

Assumptions

A complete 52-week training program for Jonathan

Priority order: Olympic weightlifting performance, hypertrophy, athleticism, longevity. Two flat barbell bench days every planned week; one targets low repetitions.

Assumption / confirmed constraint	Application
Failure scope - confirmed	Every conventional strength and hypertrophy WORK set ends at strict-form failure. Olympic lifts and their explosive teaching derivatives, warm-ups, sprints, jumps and mobility remain quality-limited. The operational definition is on page 2.
Your starting point	18-year-old male; 175 lb; 5 ft 9 in; four years of football strength training and power cleans, with less dedicated full-lift practice.
Reported best lifts, in pounds	Snatch 155; clean & jerk 205; power clean 255; back squat 385; front squat 315; bench 285; trap-bar deadlift 455. Treat these as provisional recent anchors, not proof of current daily capacity.
Unknown measurements	No conventional deadlift, strict press, full-clean or separate rack-jerk maximum is assumed. No exercise is prescribed from an invented maximum.
Health, equipment and technique	Assume pain-free training, a full gym, bumper plates and a platform. Barbell failure work requires correctly set safeties and competent spotting as described on page 2. A coach can check receiving and release; a no-coach regression is included.
Availability and schedule	No time ceiling. Four weightlifting sessions; failure work follows B/D, normally Tuesday/Friday, optionally in a second visit. Whole sessions can roll around events (page 23); bench exceptions use pages 19-20. No competition date or weight class supplied.
Physique	Side delts and upper chest receive the greatest discretionary hypertrophy emphasis. Traps also receive an early dose trial in the first successful cycle (page 15). No glute isolation; retain useful full-depth squats.
Pickup and alcohol	Pickup basketball/football zero or about once weekly, unpredictably. Alcohol usually Saturday, sometimes Friday. Training adjustments cover the day of and after; neither event is assumed harmless.
Evidence and scope	Targeted peer-reviewed review through 13 September 2026, including fresh exercise comparisons. Workout guidance only. Exact doses and scheduling are reasoned inference, not a tested optimal combined program. New selection/trial details: page 34.

S = strong evidence for the stated finding; PE = partial evidence; RI = reasoned inference. These are narrative labels, not formal GRADE ratings. Direct trial findings and exact applications are separated on page 33. Your failure preference is a binding design constraint, not a finding that failure is superior. The plan seeks the best recoverable training within it; it cannot promise the same weightlifting outcome as an otherwise identical non-failure plan.

Exactly what failure means

Failure is the endpoint of a defined task. Keep range, position and rep standard fixed throughout each set. Use full comfortable ROM by default; only the preplanned page-34 calf trial uses a partial range. **Operational rules: RI; terminology/fatigue evidence [49,50,52,53].**

Work type	Endpoint and execution
Stable machine / cable work	Continue until another complete concentric repetition cannot be achieved with the prescribed form. One unsuccessful controlled attempt may end the set. Return the resistance safely; no forced repetitions, drop sets, rest-pause extensions or assisted negatives.
Free-weight strength work	Reach the last possible complete repetition with the prescribed form: no further valid rep in reserve. Re-rack when another attempt would require altered position or unsafe loss of control. If a rep stalls unexpectedly, the spotter/safeties end it immediately. Do not deliberately seek collapse.
Technical limitation comes first	If position fails before the target muscle does, stop and record TECH rather than claiming muscular failure. Adjust load, exercise or setup next exposure. Pain, dizziness and loss of control are stop conditions, not successful failure endpoints.
Barbell bench	Use a full-thumb grip, stable feet/upper back and controlled chest touch with a brief pause. Set safeties to permit your normal range yet catch the bar if your arch relaxes; test with an empty bar. Use a competent spotter, especially on the low-rep day. No unprotected failed rep.
Front / back squat	Use a rack with safeties just below your controlled bottom position and verify a safe exit with a light bar. End at the last repetition with intact depth, bracing and balance. Front-rack collapse or a folding back is technical termination, not a rep to fight through.
Olympic and explosive work	Snatch, clean & jerk, hang lifts, rack jerks and explosive snatch pulls use the stated short sets and technical-effort ceilings. Never extend them to muscular failure. A miss triggers the quality rules, not another repetition to complete a quota.
Warm-ups / mobility / athletics	Preparation sets stay easy; jumps and sprints stop for quality loss; stretches stop at mild-to-moderate tension. These are explicitly outside your confirmed failure requirement.

Rep ranges choose the load; they do not replace the endpoint. A 3-5-rep failure set means selecting a load expected to permit 3-5 valid reps. If six are possible, do not falsely log failure at five; log the actual outcome and adjust the next exposure. Do not add another failure test to repair a load-selection error.

Zero RIR on the final completed rep and an actually unsuccessful concentric attempt are related but not identical research conditions. The free-weight prescription uses the former when a failed attempt would be unsafe. No reserve target of 1-3 RIR is hidden inside conventional work sets.

What the evidence changes

Evidence and applicability	Programming consequence
S: resistance training works	Broad evidence supports resistance training for strength and muscle growth [1]. This does not establish an optimal dose for a trained weightlifter also doing every work set to failure.
PE: failure is not required	Older syntheses [55-58] and the 2026 review [10] do not establish failure as universally superior. The latter found a small dynamic-strength advantage for non-failure and no statistically clear hypertrophy/power difference. Your preference remains binding.
PE: hypertrophy findings differ	Closer estimated proximity is associated with more growth [9]. Refalo 2024 [50] found similar quadriceps growth at failure and 1-2 RIR; Hermann 2025 [51] found modest advantages for some outcomes in single-set training. Definitions, doses and populations differ.
PE: fatigue and velocity loss	Failure can increase acute fatigue [49,52]. Velocity-loss trials [48,59,60] show different strength, hypertrophy and athletic adaptations, but velocity loss is not an interchangeable measure of failure. No study supplies a universal recovery time or set-reduction percentage.
S: volume trend / PE: heavy practice	More weekly work has diminishing returns [8]. In junior weightlifters, a middle amount of very heavy work outperformed the lowest/highest amounts on some outcomes [3]; moderate total volume also performed well [2]. Neither minimal heavy practice nor frequent maxing is established as best.
RI: the combined prescription	Four Olympic days, two failure days, the annual phases and every exact dose are individually monitored design choices. Studies have not tested this combination or proved it has zero cost to weightlifting performance.

Use the hierarchy to make decisions

Keep productive Olympic practice and the strength work that supports it first. Develop lagging muscles next, then athletic qualities, then health-only conditioning. Remove an identifiable interfering addition immediately; otherwise reverse lower-priority additions before cutting useful higher-priority work. **RI.**

Research boundary: 13 September 2026; this is a targeted review, not a registered systematic review. The ACSM overview [1] searched through October 2024; newer studies are assessed separately. Page 33 separates direct findings from individual applications; page 34 records exercise comparisons. One-set failure research [51] has no multiple-set comparator; it cannot establish dose optimality.

Weekly rhythm and session order

Day	Required work, in order	Lower-priority opportunity
Mon / A	Warm-up; snatch; easy clean & jerk.	Established athletic module after A, preferably a separate visit; pages 21 and 29.
Tue / B	Warm-up; clean & jerk; hang snatch; front squat; low-rep flat bench; upper block; lower block.	First failure day. Protect quality and full rest; split visits as below.
Wed	Eligible easy aerobic work; targeted mobility.	Normal calendar: bench relocation only. A deferred whole B/D session may roll here (page 23).
Thu / C	Warm-up; snatch; independently loaded rack jerk; explosive snatch pull.	Athletic relocation slot if Monday was missed; initially not a second module.
Fri / D	Warm-up; snatch; clean & jerk; back squat; moderate-rep flat bench; upper block; lower block.	Weekend normally lifting-free; bench exceptions on pages 19-20.
Sat	Eligible easy aerobic work and mobility, or rest.	Pickup/alcohol rules depend on actual exposure and recovery.
Sun	Off or targeted mobility; optional easy walking.	Return to A ready; no missed-volume debt.

Split at the priority muscles - RI

Preferred split when practical: visit 1 = Olympic work, squat, flat bench, incline and lateral raises (9 base failure sets). Visit 2, at least 3 h later = re-warm, upper rows 3-8, then lower block (13 sets). A single visit is acceptable while execution and full rests hold. The split changes neither weekly sets nor bench frequency; no recovery guarantee [4,11,49,52].

Compare later-exercise first sets with their usual loads, session position and reps. Repeated loss of 2+ reps or poor execution despite full rest: trial the split for two exposures. If impairment persists, remove the latest addition or use pages 14/18. Address obvious impairment immediately. Day names are the normal calendar; whole-session deferral rolls the sequence (page 23). Omitted blocks from a completed session are not replayed; only bench is rescued separately (page 20). **RI.**

Normal bench spacing is Tuesday-Friday about 72 h, then Friday-Tuesday about 96 h. Preserve at least 48 actual hours when shifting, preferably the normal spacing. This is a scheduling guardrail, not a recovery test. Keep bench after Olympic work and avoid relocating it immediately before a vulnerable priority session; page 20.

Load anchors and the entry weeks

SN = 155 lb; CJ = 205 lb initially. Snatch variants and explosive snatch pulls use SN; full clean & jerk uses CJ. Rack jerk receives its own demonstrated technical reference (RJ) and progressive working loads on pages 27-28. Until assessed, use only secure light technique practice. Do not use the 255-lb power clean to load full clean & jerks. Use a lower secure reference if the reported lift cannot currently be reproduced with sound technique. **RI; [7,44-46].**

%	SN load (lb)	CJ load (lb)
50	75	100
60	90	120
65	100	130
70	105	140
75	115	150
80	120	160
85	130	170
90	135	180
95	145	190

This chart rounds down to 5-lb totals. Use smaller change plates where available; keep units consistent. For failure work, use the actual rep endpoint rather than a fixed percentage/RPE conversion. **RI.**

First exposure only	Starting estimate and adjustment
Low-rep flat bench	Target 3-5 valid reps. About 245-250 lb is only an initial estimate from the reported 285; select lower if warm-ups show that max is stale. No preliminary max test.
Front / back squat	Target 4-6 valid reps. Front squat 260-265 lb is provisional. Back squat 320-325 lb is an estimate ONLY if the reported 385 used the same high-bar depth/setup. Otherwise use a conservative familiar high-bar load; do not convert an unknown squat style.
Moderate bench / accessories	Choose a familiar load expected to reach failure inside the row's range. Warm up first. If no history exists, make a conservative estimate, log the result, and adjust next time.

First cycle only: build tolerance without changing effort

Week 1: bottom of Olympic ranges, two-thirds of Olympic sets rounded up. One set of each upper/lower-block exercise per Tuesday/Friday; squat and bench remain one each. **Week 2:** full Olympic sets if green; upper block incline 2, lateral raise 3, row 2, all other rows 1; lower block all rows 1. **Week 3:** full base blocks only if both prior weeks were green. All conventional work sets still reach the page-2 endpoint. Introduce athletics only after two stable weeks without another workload increase; normally week 5 after the entry ramp, earlier only if the chosen lower starting dose was held stable. **RI.**

Never keep a heavy number because it appears here. Unexpectedly fewer than 3 low-rep bench reps or 4 squat reps means the load estimate needs review; use safe termination and adjust next exposure. Extra calibration sets are not owed.

The 52-week progression

Repeat this 13-week cycle four times. Interruptions may extend elapsed time; do not compress sessions to finish on a date. Structured variation has strength evidence [6]; this exact annual sequence is **RI**.

Cycle week	Year weeks	Prescription
1-3	1-3 / 14-16 / 27-29 / 40-42	Foundation (F): technical repeatability and base failure work. Entry ramp only on first onboarding or after a long interruption. Establish clean/RJ references when eligible.
4	4 / 17 / 30 / 43	Green checkpoint: repeat week 3, no additions. Exceptions: page-20 load corrections and page-27 assessment replacements. Fatigue flags use page 18.
5-7	5-7 / 18-20 / 31-33 / 44-46	Build (B): progressive heavier practice and tolerated failure work. Squats 3-5 reps. One controlled dose trial at a time.
8	8 / 21 / 34 / 47	Green checkpoint: repeat week 7, no additions. Same page-20/27 exceptions as week 4; reduce only when the review calls for it.
9-10	9-10 / 22-23 / 35-36 / 48-49	Realization (R): specific singles. Retain both squats and productive base hypertrophy; hold tolerated additions steady. No new lower-priority volume.
11	11 / 24 / 37 / 50	R continues. Tuesday normal tolerated dose; Friday one failure set per prescribed conventional exercise, no added sets. Last planned lower failure work is Friday, seven days before testing.
12	12 / 25 / 38 / 51	True taper and Friday mock meet. Low-rep bench Monday; moderate bench Saturday. Other conventional work omitted. Page 19.
13	13 / 26 / 39 / 52	Pivot: lighter Olympic practice with both squats/bench and base hypertrophy retained if recovered. If residual fatigue, page 18 gives the smaller alternative.

Phase gates: enter B after two normal weeks with at least 90% technically acceptable Olympic work reps and no recurring receiving limitation. Enter R when 80-85% singles are secure. Otherwise repeat the last successful phase in those slots; use a technical benchmark at no more than 85% in week 12. Review checkpoints still occur. **RI**.

At each pivot, review demonstrated lifts, technique, failure-set performance and physique. Change one input at a time. More volume is tested rather than automatically awarded; less volume is not imposed just because four weeks have passed. Pages 27-33 cover component progression, monitoring, overhead assistance and evidence applications.

Olympic phase prescriptions

Quality-limited work: never failure. A 1+1 = full clean then jerk. Use SN/CJ anchors; RJ has its own reference (page 27). Rest/execution: pages 10-12. Approved page-28 replacements and page-34 pull omission override the base rows. Exact doses **RI** [2,3,33,46].

Session / exercise	Foundation	Build	Realization
A / snatch	6 x 2, 65-75% SN	6 x 2, 70-80% SN	5 x 1, 78-85% SN
A / clean & jerk	3 x (1+1), 60-70% CJ	3 x (1+1), 65-75% CJ	3 x (1+1), 70-78% CJ
B / clean & jerk	6 x (1+1), 65-75% CJ	6 x (1+1), 72-82% CJ	5 x (1+1), 78-85% CJ
B / hang snatch	3 x 2, 55-65% SN	3 x 2, 60-70% SN	3 x 1, 65-75% SN
C / snatch	6 x 1, 70-80% SN	5 x 1, 75-85% SN	4 x 1, 75-82% SN
C / rack jerk	3 x 2, initially 75-85% RJ	4 x 1, initially 85-95% RJ	3 x 1, initially 90-100% RJ
C / explosive snatch pull	2 x 3, 90-100% SN	2 x 3, 95-105% SN	Omit
D / snatch	4 x 1, 75-85% SN; page 28 allows progression	4 x 1: 80, 85, 88, 88-90%; heavier only via page 28	4 x 1: 80, 85, 88-90, 90%; heavier only via page 28
D / clean & jerk	4 x (1+1), 75-85% CJ; page 28 allows progression	4 pairs: 80, 85, 88, 88-90%; heavier only via page 28	4 pairs: 80, 85, 88-90, 90%; heavier only via page 28

Effort: ordinary F work at most 7 except D at most 8. Eligible page-27 component assessment singles may reach 8 in F/B, replacing the ordinary cap only for those singles. B/R ordinarily at most 8; pulls at most 7. Effort 7 means crisp/repeatable, 8 challenging but securely controlled, not RIR. The final D single may approach 9 only in B/R through page 28; additional heavy replacements stay <=8. Secure positions, no grind/press-out. **RI**.

Full snatch/CJ work above 90% outside test week requires page 28. The final D slot can earn 90-92% in B/R, then 93-95% in R. An additional heavy slot goes first to final C snatch/B CJ, then penultimate D; the base third D slot stays <=90% until earned. If any step is unready, repeat the last secure lower load within the planned attempts. First onboarding starts low; later F may retain a secure heavier D exposure. RJ bands are starting guides; page 28 permits direct progression. **RI**.

Normal F totals 28 full/hang snatch reps, 13 full cleans and 19 jerks, plus six pull reps. Assessment replaces work rather than adding it. Misses count. Added heavy attempts replace planned attempts and remain bounded by page 28; no missed-rep debt or weekly max requirement.

General warm-up and work-set preparation

Every warm-up remains submaximal. You confirmed that preparation work is excluded from the failure requirement. Exact sequences are **RI**; warm-up evidence [31,32].

Order	Dose	Intent
1 / Raise temperature	4-6 min easy cycle or brisk walk, RPE 2-3/10.	Finish warm, without leg burning.
2 / Move	One round: ankle rocks 8/side; bodyweight squat 6; reverse lunge 4/side; thoracic rotation 5/side.	Smooth comfortable range; no long loaded stretches.
3 / Prepare upper body	Wrist circles 10 each direction; wall slide 8; very light band external rotation 10/side.	No fatigue or forced range.
4 / Specific lifting	Use page 9 for each relevant Olympic lift, squat and bench.	The second Olympic lift still needs its own abbreviated ramp.
5 / Upper and lower blocks	Before the first incline press: about 40% work weight x 8, then 65% x 4. Before row/pulldown: one easy 6-8-rep set. Before each new isolation: one easy 6-8-rep set.	Rest 60-90 s; use an additional ramp only if needed. No failure warm-up to discover the work weight.

When splitting sessions

Before visit 2: 3-5 min easy movement, then the local ramps above. If idle for more than 15 minutes within a visit, use 2 minutes easy movement and two brief ascending rehearsals before challenging work. This is a preparation rule, not a proven recovery interval. **RI**.

Tempo and rest conventions

Conventional work: approximately 2 seconds lowering, controlled transition and intent to lift promptly; a slow final rep is expected, but altered form is not. Calves pause 1 second at the bottom; bench pauses briefly on the chest. Do not add deliberately prolonged negatives or stretch holds to intensify failure. **RI**; [11,14-16,49].

Rest is time after the set ends. Use 4-5 min for squat/low-rep bench, 3-4 min for moderate bench, incline and compound pulling, and 2-3 min for isolations. Longer is allowed if required to recover performance; a timer never forces an unready set. Exact durations are **RI** informed by rest and fatigue evidence [11,49,52].

Warm-ups can reveal that a failure set should be omitted today. Do not use a failed warm-up as proof of readiness, and do not relabel an abandoned unsafe work set as a completed training target.

Lift-specific warm-ups

Lift	Light-bar sequence	Ramp before work
Snatch	Snatch-grip RDL 5; muscle snatch 3; overhead squat 3 with 2 s pause; high-hang full snatch 3.	40% SN x 3; 50% x 2; 60% x 1. Add 70% for work at 75% or more, and 80% for work at 85% or more. Rest 45-90 s early, 2 min late.
Clean & jerk	Clean-grip RDL 5; tall/light clean 3; front squat 3; jerk dip 3; unweighted split footwork 3; light jerk 3.	40% CJ: 2 cleans + 2 jerks; 50%: 1+1; 60%: 1+1; add 70/80% as above. Rest 1-2 min early, 2-3 min late.
Rack jerk after snatch	Front-rack hold 10 s; dip 3; unweighted split footwork 3.	Ordinary work: 40% of planned WORK load x 3; 60% x 2; 80% x 1. Skip stages at/above work. Rest 1-2 min early, 2-3 min late. Assessment uses page 27 instead.
Explosive snatch pull	Three controlled floor starts with about 60% of work weight.	One further light set if positions need rehearsal; never test pulling failure.
Front / back squat	Empty bar x 5; verify rack safeties and depth.	About 40% work weight x 5; 60% x 3; 80% x 1; 90% x 1 when useful. Rest 1-2 min early; 3 min before work.
Flat bench, both days	Empty bar x 10; verify safety height, grip and chest touch.	About 40% work weight x 6; 60% x 3; 80% x 1; 90% x 1 before low-rep work. Rest 1-2 min early, 3 min before work.

Skip any Olympic ramp stage that would reach or exceed a light work load. If the standard bar distorts positions, use a lighter technique bar. No ramp is a rep-max test. All exact ramps are **RI**; [31-33].

Receiving and release check

Show three secure light overhead squats, front-rack squats and split catches without pain. If a position is not secure, use page 16's regression. Learn release of a missed snatch/jerk/clean on a clear platform from a qualified coach before challenging catches. Without that instruction, remain with secure technique-bar/light practice. Keep catches unstrapped. **RI**.

A belt is optional on familiar squats and heavier clean & jerks if it does not obstruct positions. Straps are optional on explosive pulls and supported shrugs, not on learning catches. Equipment does not change the stop rules.

Monday A and Thursday C

Complete the warm-ups. A/C normally contain no conventional failure work; test-week Monday bench and bench relocations are exceptions (pages 19-20). A deferred whole B/D session may fall on another weekday under page 23. Foundation rows are below; page 7 governs B/R.

Monday, in order	Work / intensity	Rest / execution
1 / Full snatch	6 x 2, 65-75% SN; effort at most 7.	2-3 min. Reset on floor 10-20 s between reps; balanced full receive.
2 / Full clean & jerk	3 x (1+1), 60-70% CJ; effort at most 7.	2-3 min. Stand clean securely, then jerk; hold stable before recovery.
Thursday, in order	Work / intensity	Rest / execution
1 / Full snatch	6 x 1, 70-80% SN; effort at most 7.	2-3 min. Same technical standard at every load.
2 / Rack split jerk	3 x 2, initially 75-85% RJ; effort at most 7. Assess/progress on pages 27-28.	2-3 min. Reset safely 10-20 s between reps, or two singles 30 s apart if returning to rack is awkward.
3 / Explosive snatch pull	Provisional 2 x 3, 90-100% SN; effort at most 7. Review: page 34.	2-3 min. Match snatch start/extension; reset each rep. Stop if speed/position deteriorates; never muscular failure.

After A: the eligible page-21 athletic module. After C: relocation only at first; page 29 governs expansion. Failure work belongs to B/D, normally Tuesday/Friday; whole-session shifts and separate bench exceptions follow pages 19-20/23. No replay of omitted accessories. **RI**.

Why normal A/C days stay free of failure work

Tuesday and Friday need high-quality Olympic practice before their failure blocks. Avoiding hard conventional work the preceding day is a scheduling choice, not proof of a universal recovery period. Monday and Thursday also reveal whether the prior failure dose is tolerable. **PE/RI; [4,49,52]**.

Frequent short sets provide full-lift rehearsal while limiting repeated deteriorating catches. Moderate-volume weightlifting evidence supports restraint, but does not prove these exact rep counts are optimal. **PE/RI; [2,3,33]**.

If Monday doubles repeatedly compromise Tuesday, reduce Monday snatch to 4 x 2 before adding work elsewhere. If Thursday overhead positions suffer after Tuesday pressing, remove the interfering pressing dose or use page 17; do not lower the standard for a successful jerk.

C-to-D recovery check: compare Friday CJ at familiar loads after comparable snatch work. If a Thursday jerk increase precedes repeated poorer Friday lifting, reverse that increase first. If still impaired, trial one fewer C jerk set for two exposures, suspending C assistance; retain only if Friday improves and ordinary jerk execution stays normal. Check any Thursday athletic addition too. If the cause is unclear, follow page 17's priority order. Immediate impairment uses page 17. These observations do not prove causation. **RI [3,33]**.

Tuesday B: priority work, then failure

Order / exercise	Work / intensity	Rest / execution
1 / Full clean & jerk	6 x (1+1), 65-75% CJ; technical effort at most 7.	2-3 min. Secure clean, deliberate reset, stable split jerk; no failure extension.
2 / Above-knee hang full snatch	3 x 2, 55-65% SN; target effort 6-7, ceiling 7.	2 min. Stand, hinge to just above knee, pause 1 s, lift into a full receive. Reset between reps.
3 / Front squat	1 work set to strict-form failure; target 4-6 reps.	4-5 min before the next hard lift. About 2 s down, full controlled depth, fast intent up; safe endpoint on page 2.
4 / Flat barbell bench: low-rep	1 work set to strict-form failure; target 3-5 reps.	4-5 min before other pressing. About 2 s down, brief chest pause, full lockout; spotter and safeties.
5 / Upper block	Page 13, in its listed order and phase dose.	Split after lateral raises under page 4; re-warm before resuming.
6 / Lower block	Page 14, in its listed order and phase dose.	Only after upper block; lagging-muscle work gets priority within hypertrophy.

The strength dose is deliberately small at entry

One front-squat set and one back-squat set are entry doses. In 32 trained men, eight squat sets to failure outperformed one for squat strength [74]; that does not prescribe eight sets here. Failure does not replace volume. At the first normal F/B review after onboarding and two stable green weeks, explicitly consider one extra relevant squat set under pages 28/31. Record why you trial or hold it; no plateau is required. Exact dose **RI**.

B phase: front squat targets 3-5 reps at failure. R phase: both front and back squat use one 3-5-rep failure set on their normal days; keep useful strength support unless its recovery cost requires reduction. Bench stays one 3-5-rep failure set in normal F/B/R weeks. Reduced and test weeks use pages 18-19. Exact changes are **RI**.

A failed jerk is not a successful clean & jerk pair. If the Olympic session is amber or red, its fatigue rule can remove the entire failure block; the written strength rows are not permission to train through impaired receiving or coordination.

Friday D: total practice, then failure

Order / exercise	Work / intensity	Rest / execution
1 / Full snatch	4 x 1, 75-85% SN; effort at most 8.	3-4 min. Same load or small steps inside range. No extra heavy attempts to repair misses.
2 / Full clean & jerk	4 x (1+1), 75-85% CJ; effort at most 8.	3-4 min. Secure recovery from clean; hold jerk overhead before controlled foot recovery.
3 / High-bar back squat	1 set to strict-form failure; target 4-6 reps.	4-5 min before next hard lift. About 2 s down, full comfortable depth; do not fight a folding torso.
4 / Flat barbell bench: moderate	1 set to strict-form failure; target 6-8 reps.	3-4 min before other pressing. Same pause and setup as Tuesday; load will normally be lower.
5 / Upper block	Page 13, in its listed order and phase dose.	Split after lateral raises under page 4; re-warm before resuming.
6 / Lower block	Page 14, in its listed order and phase dose.	No additional finishers.

B phase: squat targets 3-5 reps at failure. R phase: retain one 3-5-rep back-squat set and tolerated base blocks.

Week 11 Friday: one failure set per prescribed exercise, including squat and bench; no added sets. All conventional work retained in any phase still uses failure. **RI**.

Why Friday ends the failure week

The normally lifting-free weekend gives more scheduling space before Monday than a routine Saturday upper-body session would. Test-week and relocated bench are exceptions (pages 19-20). Neither spacing nor familiar exercise proves that recovery is complete. **PE/RI; [29,30,49,52]**.

Record the best secure Olympic singles before doing failure work. Compare them with like-load performance across normal weeks, then check Monday/Thursday receiving quality. An improving squat rep maximum is useful, but cannot justify worsening the classical lifts. **RI; [44,45]**.

Page 24 adjusts for the actual alcohol exposure and readiness. Do not automatically move Friday failure work onto Thursday before priority lifting. Defer only when needed and preserve bench spacing.

Upper block: Tuesday and Friday

Perform after Olympic work, squat and flat bench. In the preferred split, rows 1-2 finish visit 1; rows 3-8 begin visit 2 after re-warming (page 4). These are **per-session** base sets in normal F/B/R weeks. Entry ramp: page 5; checkpoint, week-11, taper and pivot overrides: pages 18-19. Warm-ups never go to failure.

Order / exercise	Sets x target reps	Rest	Execution
1 / Incline machine press, 30-45 degrees	3 x 6-10	3-4 min	Choose a comfortable angle and stable deep range; keep the setup consistent. About 2 s down; safe stop. Incline Smith/DB alternative: page 25.
2 / Cable or DB lateral raise	4 x 12-20	2-3 min	Slightly forward arm plane, raise to about shoulder height; 2 s lowering. No torso swing or shortened-ROM finishers.
3 / Chest-supported row	2 x 8-12	3 min	Chest stays on support; full comfortable reach; no trunk heave.
4 / Lat pulldown	1 x 8-12	3 min	Controlled overhead reach, pull to upper chest; no swing or behind-neck pull.
5 / Supported machine or DB shrug	1 x 10-15	2-3 min	Full comfortable elevation; controlled lowering. Straps allowed; no neck circles or forced stretch.
6 / Reverse pec deck	1 x 12-20	2-3 min	Stable torso; comfortable rear-delt range; no lumbar extension.
7 / Cable curl	1 x 8-12	2-3 min	Supinated grip; upper arms beside torso, shoulders neutral. Full comfortable elbow range; no torso assist.
8 / Overhead cable triceps extension	1 x 10-15	2-3 min	Upper arms comfortably overhead and stable; controlled deep elbow bend, then extension. About 2 s return; no torso heave. Fallback: page 34.

Incline evidence used 44-degree Smith pressing; only 30 novices supplied hypertrophy data [13]. It does not prove an optimal angle. Cable/DB raises produced similar growth in 24 trained adults [12]. Overhead extensions are preferred from the direct triceps trial [69]; exact doses remain **RI**.

Neutral cable curls remain reasonable after the matched shoulder-angle trial [73]. Rows, pulldowns, shrugs and reverse pec deck are practical selections without an established hypertrophy ranking against alternatives. Support does not prove negligible fatigue. Full ROM is the default, not a universal superiority claim [14-16]. **PE/RI**.

For multiple failure sets, keep load if later sets remain inside the target range. If the prior set fell below the lower bound despite full rest, reduce the next set's load about 5-10%. Do not change it mid-set, add sets, or chase the first set's rep count.

Lower block: Tuesday and Friday

Perform after the upper block. The priority-supporting front/back squat has already been completed before bench. These are base doses per session, not additional squat work.

Order / exercise	Sets x target reps	Rest	Execution
1 / Seated leg curl	2 x 8-12	2-3 min	Hip flexed, pelvis secured; full comfortable knee range. About 2 s return; no hip lifting.
2 / Standing calf raise, both days	2 x 10-15	2-3 min	Knees extended, stable support. About 2 s down, 1 s bottom pause, full rise. No bouncing. Planned partial-range option: page 34.
3 / Supported reclined leg extension	1 x 10-15	2-3 min	Prefer about 40 degrees hip flexion if the machine supports it securely. Pelvis/back supported, knee aligned; full comfortable knee ROM. Setup/fallback: page 34.
4 / Machine abdominal crunch	1 x 10-15	2-3 min	Controlled trunk flexion without hip pulling, neck strain or jerking; select a supported machine with a safe stop.

Why these exercises

Direct comparisons / PE: seated curls favored hamstring growth [39]; standing calf raises favored gastrocnemius growth, with no detected seated advantage for soleus [70]. Reclined extensions favored rectus femoris growth [71]; knee extension also complements squats without added glute work [72]. These studies do not validate the exact doses in this trained lifter. Machine crunch is a practical **RI** choice, not a demonstrated best abdominal exercise.

Full-depth squats still load the glutes. Zero glute isolation means no added hip thrusts, lunges or RDL accessories; it does not mean zero glute stimulus. Squat/hip-thrust hypertrophy research [40] supports taking squat contribution seriously, not assigning it a precise fractional value. **PE/RI**.

Fatigue adjustments

If leg soreness or loss of force affects the next Olympic practice, remove the added lower-block set first when it is the obvious cause. If the base block is responsible, reduce curls/calves to one set each and omit extension/crunch temporarily. Keep necessary squat support only if its own quality and recovery are acceptable. All remaining conventional sets still end at failure. **RI**.

A lighter load taken through many more repetitions to failure is not automatically a deload. Manage fatigue by fewer sets, fewer exercises or omission of the affected block; do not turn a planned 4-6-rep squat into an exhausting high-rep set.

Hypertrophy allocation and dose trials

Muscle / exercise	Normal weekly base	First review direction
Upper chest / chest	6 incline sets + 2 flat bench sets	Review incline at 8 weekly sets before adding flat sets. Flat is not counted as an equal upper-chest replacement.
Side delts	8 direct lateral-raise sets	Trial 10, then 12 if higher-priority work permits. These are review points, not scientific ceilings.
Traps	2 direct shrug sets plus Olympic work	Review 4 direct sets alongside upper chest/delt priorities. No automatic half-set credit from Olympic sets; schedule a trial during the first successful cycle.
Back / rear delts	4 rows + 2 pulldowns; 2 reverse-pec-deck sets	Progress reps/load; later add 1 weekly set if useful.
Quads / glutes	2 squat failure sets + 2 leg-extension sets; zero glute isolation	Squat support belongs to priority 1 when it improves lifting. Increase only after assessing that need and recovery.
Hamstrings / calves	4 seated-curl sets; 4 calf sets	Progress load/reps, then trial one weekly set if tolerated and desired.
Arms / trunk	2 curl + 2 overhead-extension + 2 crunch sets	Indirect work contributes; these are entry doses, not permanent maintenance caps.

Total: 44 conventional failure sets per normal F/B/R week: 22 Tuesday and 22 Friday, plus the separate Olympic work. This is a starting distribution, not a validated optimal dose or conversion from the previous program. Fewer sets do not guarantee less fatigue when effort rises. **RI; [8,49-52].**

A controlled way to look for more productive work - RI

After two normal green weeks, you may trial **one additional weekly set** even if progress has not stalled. Hold that dose for two normal weeks before deciding whether to retain it or test the next set. First resolve any priority-1 practice/strength need; then emphasize incline and side delts. Give traps a trial from 2 toward 4 weekly direct sets during the first successful cycle; do not wait to maximize both larger priorities within hypertrophy first. Add one weekly shrug set at a time, as with the other rows. Put the set at the end of its existing row; hold other training changes for two normal weeks.

Retain the addition only if like-load Olympic quality, next-session positions and failure-set performance remain healthy. More repetitions/load is useful tracking, not proof of faster muscle growth. Compare standardized photos/circumferences over a full cycle; if an addition worsens a higher priority, reverse it. Do not force all review targets in one cycle. **PE/RI; [8,41].**

Expand one domain at a time: weightlifting support first, then hypertrophy, then athleticism, then health-only aerobic work. This reduces ambiguity about fatigue; it still cannot detect every opportunity cost or prove a global optimum.

Technique: observe before prescribing

Your clean & jerk is about 80% of the power clean and 65% of the front squat. These suggest a possible gap in technical expression or jerk capacity; they do not diagnose the limiting structure or predict missing pounds. **PE/RI**; [44,45].

Observed issue	Replace, do not add	Return gate
Unsafe overhead receive or front rack	Replace affected Olympic rows with 3 x 3 technique-bar overhead/front squats, 2 s pause, 60-90 s rest, effort at most 4. These are exempt technical rehearsals.	Three secure light warm-up reps and a safe release check; return with 4 x 1 at 40-60%, rest 2 min.
Reluctant or slow turnover	Use high-hang full snatch 3 x 2 at 40-60% in B's hang slot; for clean, replace A's CJ with high-hang full clean + jerk 3 x (1+1), 40-60%.	Two exposures with at least 90% secure reps; restore ordinary variation, retaining floor practice where safe.
Balance lost floor to knee	Replace first two work sets with knee-pause singles at 50-65%, 2 s pause; rest 2 min. Remaining planned work at a secure lower load.	Repeatable balance across two exposures.
Jerk stance / recovery	Replace C rack jerk with light-bar split jerk 3 x 2, 2 s split hold, 90 s rest; first do 2 x 3 unweighted footwork reps. Effort <=6.	Stable split and secure recovery across two exposures.
Dip drift: technique regression	Replace C rack jerk with pause-dip split jerk 3 x 2 at 40-60% RJ (CJ until assessed): pause 1 s at dip bottom, drive/receive normally, hold split 2 s. Rest 90 s; effort <=6. Use less load if needed.	Two exposures with controlled vertical dip and secure receive. Page 31 is a separate, heavier assistance trial only after positions and RJ are established.

Feedback routine

Film first and last work sets from a consistent side or front-quarter angle, showing feet and bar. Self-rate first, then review one recurring error. Choose one useful cue, not an entire checklist at once. Feedback has partial support [34]; a small skilled-lifter study did not establish a universally superior cue style [35]. Seek a coach if the same error persists for two weeks. **PE/RI**.

A full-lift rep must be balanced, securely received and recovered without chasing or press-out. Do not force an artificial catch pattern solely to meet a ratio. Recover a split jerk by bringing the front foot partway back, then the rear foot forward under control. **RI** [33].

Do not convert a missed or slow Olympic set into a failure workout. The technical rehearsals above remain easy despite using squat-like positions; they are not replacements for the separately identified conventional strength work sets.

Readiness, misses and fatigue

State	Olympic work today	Conventional failure work
Green	Normal pain-free positions and coordination; usual warm-up loads look/feel normal. Use phase prescription.	Use prescribed phase sets if local warm-ups also pass.
Amber: global fatigue or unusual heaviness	Rest 3 min; reduce load 5-10%; do at most two-thirds of planned sets, rounded down, minimum 1 and never more than planned. Effort at most 7.	Omit failure work today. Defer bench within the spacing rule if possible; do not replace it with an easier high-rep failure set.
Amber: isolated local problem	Keep only Olympic movements unaffected by it. Regress/omit affected movement.	Omit the affected exercise/block. Unaffected work may use one set per row only if normal control and local warm-ups are clear.
Reduced Olympic work still poor	At most four easy singles at 50-60%, capped by remaining planned reps, only if secure; otherwise stop.	No failure work.
Red	Pain changes movement, acute instability, dizziness, intoxication or unsafe release: stop loaded/explosive training.	No failure work. Persistent/worsening pain or suspected injury warrants appropriate assessment before resuming.

During an Olympic lift: two consecutive misses or repetitions with the same material fault: rest, drop 5-10%. If either of the next two attempts is still poor, end loaded work for that lift. Misses count; do not repay them. All thresholds and actions are **RI**; [3,7,47].

Weekly intervention

Review when two comparable sessions need over 5% less load for similar quality, good Olympic reps fall below 90% twice, or soreness changes a position twice. Remove the obvious cause immediately. If unclear, remove health-only aerobic additions, then formal athletics, then recent or locally interfering hypertrophy work. Reduce failure-set count before reducing essential Olympic practice. If the next exposure is still impaired, use the targeted or full reduction on page 18. **RI**.

Track first-set failure reps at a familiar load. A fall of more than 20% on two exposures is a review flag, not a diagnosis; illness, technique and equipment can explain changes. Do not use fixed CMJ or wearable cutoffs as proof of safe lifting. **RI** [47,49,52].

With persistent bench-related overhead impairment despite one work set on each of two days and removal of other aggravating pressing, defer the affected bench exposure. There is no honest way to guarantee both continued failure exposure and zero weightlifting cost. Pain/unsafe fatigue overrides the quota; do not compensate with consecutive-day benching.

Checkpoints, reductions and pivots

Every retained conventional work set still reaches strict-form failure. Keep the normal rep ranges/rests; reduce counts rather than prescribing hidden reserve. Exact policy **RI**; deload evidence uses different populations/protocols [36-38].

Situation	Olympic and strength work	Upper / lower dose per Tue-Fri
Green weeks 4 / 8	Repeat preceding F/B loads/sets; keep base FS/BS, earned support and both bench slots. No additions. Eligible page-27 assessments alone may replace the held Olympic rows.	Retain tolerated work. Calendar review does not automatically delete hypertrophy or athletics.
Normal R weeks 9-10	R classics; Tue FS and Fri BS each base 1 x 3-5 failure. Tolerated support trials may continue under pages 31-32. Bench Tue 1 x 3-5, Fri 1 x 6-8 failure.	Full base blocks; retain successful additions. A continuing press trial replaces one Friday incline set. No new lower-priority additions.
Week 11	R classics; Tuesday normal dose. Friday squat 1 x 3-5 and bench 1 x 6-8 failure.	Tuesday normal. Friday 1 failure set per base upper/lower exercise; no added sets. Active press substitution: page 32.
Targeted reduced week	Reduce only affected work first. If fatigue is broad but warm-ups are now green: half Olympic sets rounded up, low end of preceding phase, effort ≤ 7 ; omit pulls. Retain one FS, BS and each bench if locally ready.	Per session: incline 2, lateral 2; every other upper row 1. Lower: each row 1. Never exceed the dose actually being performed. No athletics or new doses if fatigue caused the reduction.
Full fatigue reset	If global impairment persists, A-D half sets rounded up at 50-65%, effort ≤ 6 , only while secure; omit pulls/squats. Each bench slot 1 set only if fully ready, otherwise defer.	Omit accessory blocks and formal athletics. This is a reactive fallback, not the default every fourth week.
Week 13 pivot	Replace all ordinary Olympic rows: A/C each SN 4 x 1 and CJ 3 pairs, 60-70%, effort ≤ 6 , rest 2 min; no rack jerk or pulls. B/D: no classics; squat 4-6 reps, then bench. Hold support additions until next F.	If recovered, normal base blocks. If residual fatigue but local warm-ups green, 1 set per base row. If amber/red, readiness rules override. Resume the established athletic dose if recovered; page 29.

A reactive reduction replaces the current week. After normal performance returns, repeat the last successful phase week and shift later dates. Restore reduced multi-set rows by one set per row per exposure, up to the prior tolerated dose, only while subsequent Olympic practice remains normal. Hold other additions. **RI**.

Apply phase first, then the stricter readiness/event rule: a reduction never adds sets or restores an omitted exercise. Page-20 rep-window load corrections remain allowed in held weeks. A two-bench-set week is not completely unloaded. Lighter weights for many more failure reps do not substitute for reducing workload.

Taper and mock meet: week 12

Week 12 replaces the normal week. Week 11 Friday reduced conventional work to one set per prescribed exercise. Its squat/lower work is the last planned lower-body failure exposure, seven days before the Friday test. If your prior logs show slower recovery, omit the affected Friday lower work and use Tuesday as the last exposure. Exact timing is **RI**; the recent taper review summarizes self-reported practices rather than proving this failure-constrained taper optimal [5].

Day	Work, in order	Rest / endpoint
Monday	Warm up; snatch 3 x 1 at 75-80% SN; CJ 3 x (1+1) at 75-80% CJ; flat bench 1 x 3-5 to strict-form failure.	Olympic 2-3 min, effort at most 7; full bench ramp and 4-5 min preparation.
Tuesday	Snatch 2 x 1 and CJ 2 x (1+1), both 80-85%. No conventional work.	3 min; effort at most 8, no misses chased.
Wednesday	Off; optional 10-15 min easy walk.	No pickup or athletic module.
Thursday	Snatch 2 x 1 and CJ 2 x (1+1), 50-60%.	2 min; easy rehearsal only.
Friday	General/specific warm-ups; three scored snatch attempts, then three clean & jerk attempts. Nothing else.	3-5 min between attempts; about 8-10 min between lifts, then CJ ramp.
Saturday	Flat bench only: full warm-up, 1 x 6-8 to strict-form failure only while sober, fully ready and permitted by page 24. Otherwise defer under page 20; keep it after the Olympic test.	3-4 min preparation; no other failure, athletic or formal aerobic work.
Sunday	Off or easy mobility/walking; deferred bench only if pages 20/24 permit it.	No catch-up blocks; any deferred bench keeps its full warm-up, dose and spacing.

Attempt selection - RI

Open at about 88-92% of the current technically secure anchor. After an easy valid opener, select approximately 97-100%; after a secure second lift, try the smallest defensible PR increase. Hold or reduce expectations when the opener is difficult. A press-out, unstable receive or failed recovery is not a valid make. The three-attempt limit includes misses.

A secure test attempt may exceed the ordinary training effort ceiling. If 80-85% warm-up singles are not secure, change the test to a technical benchmark at no more than 85% and reschedule the max test after recovery. Do not perform a bench or squat max test in this week. **RI**.

The Monday failure bench is four days before Friday testing; this is a planning interval, not a promise of full recovery. Bench exposure 2 stays after the Olympic test. If the test moves, move that bench exposure after it and shift the next week's low-rep bench to preserve at least 48 h; do not stack sessions to meet calendar labels.

Progression and bench continuity

Work	Progression rule - all exact thresholds RI
Olympic lifts	Add the smallest plate increment after at least 90% good work reps across a normal week, stable positions and effort within cap. Use the phase and heavy gates on pages 7/28. Only demonstrated valid lifts update anchors; separate clean/RJ references and direct jerk progression are on pages 27-28. No power-clean or repetition-equation substitution.
Squat / low-rep bench	At the exact upper rep bound, add 2.5-5 lb after two normal successful exposures. Above the bound (e.g. 6 on a 3-5 bench set), correct load by +2.5-5 lb next eligible exposure; no second overshoot required. Both require valid failure reps, no TECH or next-session cost. Rep-window corrections are allowed in held weeks; no bonus set/single.
Other conventional work	When every set reaches or exceeds the upper target bound with the same standard, add the smallest available load next exposure. A single-set exercise uses two successful exposures before increasing. If reps fall below the lower bound, reduce about 5-10% next time.
Exercise load changes	Different machines and grip/ROM changes need a fresh estimate. Log actual successful reps, failed attempt if any, and TECH/pain termination separately. No formula can make failure reps identical each day.
Volume	Page 15: one added weekly set at a time, two normal weeks of observation; no new lower-priority addition in R, review, reduced, taper or pivot. Keep successful doses where phase/readiness permits. Extra low-priority volume never compensates for poorer Olympic practice.

Protect two bench days without creating catch-up fatigue

Normal F/B/R/review/reduced/pivot: Tuesday low-rep and Friday moderate. Test week: Monday low-rep and Saturday moderate. One failure work set per exposure is the base in every phase. Keep at least 48 hours between actual sessions, including week boundaries; favor normal 72-96-hour spacing. Always place bench after any priority lifting that day. **RI.**

For a separately missed bench slot, use the nearest ready day preserving spacing: Tuesday may move to Wednesday only if Friday is at least 48 h later; otherwise Friday becomes low-rep and Sunday moderate if ready. Shift the next Tuesday when needed. Friday may move to Saturday after event/readiness review. Other omitted blocks are dropped. A whole B/D deferral instead follows page 23 and carries its bench slot; do not repeat it on the old weekday. Never count two sets on one day as twice weekly. **RI.**

When relocation would compromise priority lifting, postpone the affected sequence rather than forcing a compressed calendar week. Unsafe fatigue or pain may prevent the quota. If this repeatedly happens at the minimum dose, the failure/frequency requirements need reconsideration; there is no validated workaround that guarantees all goals simultaneously.

Athleticism: a dependable weekly slot

Default: Monday after A, preferably a separate visit at least 3 h later, early enough for Tuesday readiness. If one visit, take a 5-min transition then the running warm-up. Thursday after C is the backup slot; one module initially. Separation is a scheduling choice, not proof of zero interference [20-22,27].

Order	First two eligible exposures	Rest / quality
Preparation	4-5 min easy walk/jog, RPE 2-3; ankle rocks and leg swings 8/side; marching 2 x 10 m; 3 x 20 m progressive runs at 50, 65, 80% effort.	Walk back; rest 90 s before work. Stop for altered gait, painful stiffness or unusual heaviness.
1 / CMJ	2 x 3 bodyweight countermovement jumps; maximal intent, controlled landing.	10 s reset; 90-120 s between sets. Stop for clearly falling height or altered landing.
2 / Acceleration	3 x 10 m standing-start runs at 85-90% perceived best effort.	2-3 min between runs. Smooth acceleration, no conditioning finisher.
Acceptance	At the next B/D session compare familiar-load Olympic quality and first squat/bench failure-set output.	Suspend/reduce an addition that impairs a higher priority; check the next comparable session.

Introduction and development - RI

Introduce after two green weeks at a stable workload, with no other dose increase in the introduction week. Weeks 3 and 4 can qualify when week 3 begins the full base dose and week 4 repeats it successfully; week 5 is conditional, not compulsory. If already adapted, use the next eligible Monday. Do not repeat onboarding each cycle or wait for hypertrophy to stall.

The first two exposures familiarize; six jumps is not the long-term target. Next use **3 x 3 jumps**, runs unchanged. After two successful exposures at each new dose, change one component: runs to 3 x 15 m at the same effort, then 90-95% at that distance. Next add one 3-jump set; thereafter alternate a run step (to 20 m, then +one run) and a jump step (+one 3-jump set). Keep all rests and stop rules.

Review at 4 x 3 jumps and 4 x 20 m. Use page 29 for later options and a second exposure. These are review points, not mandatory targets or scientific ceilings. Green checkpoints retain the established dose; realization does not automatically return it to six jumps.

Plyometric benefits have broad support [21]. The basketball study [63] reports conflicting jump totals across its abstract and methods; it is not a numerical dose basis here. Six jumps/30 m is familiarization, not a proven development minimum. Warm-up potentiation is uncertain [64]. Stop for pain/instability; two comparable runs >3% slower trigger a stop with reliable timing (RI).

Aerobic work and mobility

Add conditioning only after two green lifting weeks, in a week with no failure-set or athletic-dose increase. Start in a normal F/B week after the entry ramp; the exact start date follows readiness, not a compulsory calendar. **RI**; [19,20].

Work	Starting dose	Expansion / limits
Aerobic: Wednesday and Saturday	20 min easy cycle or brisk walk each day, RPE 3-4/10, full-sentence talk test. Apply event/readiness rules; no routine Saturday athletics.	After two green weeks, add 5 min to one session, up to 2 x 30 min. Then add 10 min/week after each two green weeks toward 150 actual moderate min; split or extend walks.
Further health progression	After 150 min is well tolerated for a full cycle, further 10-min increments toward 300 are permitted.	Count actual moving minutes: 1 vigorous minute is about 2 moderate for guideline accounting. Gym visits and standing time do not count. This is not a fatigue equivalence; higher-priority expansion comes first.
Targeted mobility	Choose at most two restrictions below. After lifting or on non-lifting days: 2 x 30 s/side, rest 15 s, three days/week.	Mild-to-moderate tension, not failure or pain. Follow with five slow active reps. After two weeks without improvement, use 45-s holds or a fourth day.

Restriction	Drill / check
Ankle	Bent-knee calf stretch facing wall, heel down. Recheck balanced squat/catch without a mobility-driven heel rise.
Overhead shoulder	Hands-on-bench lat stretch with ribs controlled. Recheck a secure light overhead position.
Front rack	Gentle supported wrist-extension stretch or unloaded front-rack elbow lifts. Never force range with a heavy bar.
Hip rotation	Supported 90/90 position, small lean at the hip. Recheck comfortable receiving depth.

Held et al. (2026 [19]) is reassuring about concurrent adaptations overall; Schumann et al. (2022 [20]) identified explosive-strength concerns, particularly within one session. Neither guarantees zero interference in a weightlifter doing failure work. WHO activity benchmarks [25] and mortality associations [26] inform gradual health expansion, not an exact personal optimum. **PE/RI**.

At each dose review, a justified Olympic or hypertrophy trial comes first; then athletic progression, then health-only aerobic expansion. If added aerobic work makes the athletic slot repeatedly fail, reverse that expansion. **RI**.

Aerobic phase rules - RI: green checkpoints and R weeks 9-11 hold established minutes, no expansion. Week 12 suspends formal aerobic sessions and added brisk walks; only page-19 easy walking/mobility. Week 13 resumes the pre-taper dose if ready, without increases. Active event/fatigue restrictions override.

Walk placement - RI: use ready days without event/fatigue restrictions; on lifting days, after priority work. Example at 150 min: main Wed/Sat sessions 30 min each; additional walks Mon/Thu 20 min each and Tue/Wed/Fri/Sat/Sun 10 min each. Split longer walks if useful; count actual moderate minutes once.

S: loaded resistance training and stretching improve ROM on average [23,24]; this does not establish that more flexibility always improves lifting or prevents injury. Omit extra stretching when the relevant positions are already comfortable. No routine HIIT is added merely to save time.

Pickup sport: adjust for actual demand

Log active time, effort, fast running, jumps/cuts, contact and next-day symptoms. Casual shooting and repeated maximal play are different exposures. No game-minute formula equals a hypertrophy set or a controlled speed dose. Exact rules are **RI**; applicability of sport/recovery studies is **PE [27,28,67,68]**.

Situation	Training decision
Casual skills, easy movement	If no meaningful fast running/jumping/contact or residual symptoms, keep the normal gym and athletic plan. The name basketball/football does not trigger a reduction.
Demanding game planned later	Keep ready Olympic work first. If the game would compromise conventional work or the next Olympic day, shorten/skip the optional game. If fixed, postpone the whole session or omit its affected failure block after lifting; rescue bench under page 20. A few hours do not prove recovery.
Demanding game already completed	Postpone the affected whole session to the next ready day and roll later dates. Optional secure rehearsal: up to 3 singles per lift at 50-60%, rest 2 min; this does not complete the deferred session. Otherwise off. Brief casual skills do not trigger this automatic move.
Repeated runs, jumps or cuts	No separate athletic module on game day or while residual leg fatigue persists. The game replaces the nearest overlapping athletic exposure; Monday can relocate to Thursday once ready. A possible game later in the week does not cancel Monday in advance.
Contact / collision load	Also check painful impact sites, neck/shoulder comfort and rack/overhead range. Contact can occur in either sport; assess actual impacts. Persistent pain or altered movement invokes page 17, not a preset longer waiting period.
Day after	Normal pain-free positions, coordination and familiar warm-up performance allow the scheduled prescription. Global heaviness or altered movement invokes amber/red; affected failure work is omitted. No blanket 24/48/72-hour clearance or ban.
Two demanding games / test week	Remove the secondary athletic module first; do not force another high-intensity exposure to satisfy a quota. Avoid optional demanding games in the final 5-7 days before testing; if readiness is disrupted, postpone the maximal test.

Game warm-up - RI

3-5 min easy movement; ankle rocks/leg swings 8/side; 2 x 3 controlled landings; 2 x 10 m lateral shuffles; 2 sets of 2 planned 45-degree cuts per side at 60-75%; 3 x 20 m progressive runs from 60 to 85%, walking back. Rest 60-90 s between landing/cut sets. Stop for pain. These borrow components, not a proven exact injury-prevention dose [28].

Whole-session deferral: roll A-B-rest-C-D-rest-rest, adding recovery days as needed; normally at most two consecutive Olympic days. B/D carry their prescribed failure blocks and bench to their new dates. Example: B deferred Tuesday to Wednesday makes C Friday and D Saturday, subject to readiness/events. If normal Olympic work was completed but a failure block was omitted, drop that block; only bench is rescued under page 20. Never replay completed work or compress dates. Saturday play with normal recovery need not erase A athletics; otherwise use the next ready C slot.

Alcohol: day of and day after

These rules apply in every phase, including test-week Saturday bench. Apply the phase prescription first; event rules may reduce/defer it, never restore omitted work. Record exposure, timing and impairment. No amount or waiting time certifies unaffected recovery/adaptation. Workout adjustments are **RI**; evidence application **PE** [29,30,54,65,66].

Exposure / present state	Training action
Event later; sober and normal now	Keep usual priority lifting. For a known limited event without prior recovery problems, usual failure work may proceed; no extra/new sets. Do not automatically move Friday failure work to Thursday before Friday classics.
Larger / unfamiliar event anticipated	Keep ready priority work; hold additions and make the following day flexible. If moving conventional work is necessary, follow bench spacing and the rolling sequence. Scheduling cannot protect adaptation from what happens afterward.
Intoxicated, uncertain sobriety, dizzy, nauseated or poorly coordinated	No loaded, explosive, failure or conditioning workout and no pickup. Do not use barbell warm-ups to establish sobriety.
Next day: known limited exposure, no impairment during it; now sober, symptom-free and normal	Use ordinary readiness and local warm-ups. If they pass, perform the normal scheduled session, including its failure work. No automatic day off, 80% ceiling or all-exercise reduction solely because it is the next day.
Larger, unknown or previously impairing exposure; now sober and symptom-free	Verification return: start Olympic work at the low end of the planned range, no PR or extra heavy attempts. If local checks pass, at most 1 conventional set per already prescribed row; otherwise omit. No new/progressed athletics. Restore phase volume after this return and the next warm-ups remain normal.
Any exposure with abnormal movement or warm-up performance	Apply page 17 amber/red. Omit unsafe failure work; reassess at the next scheduled session. Elapsed hours neither require a fixed ban nor guarantee readiness.
Game plus alcohol / testing	Use the stricter active restriction for the combined demands. Postpone maximal testing when recovery is uncertain; preserve bench spacing after relocation. Never compress missed work.

Separate small Barnes studies found no detected added force deficit at a lower studied dose [65], but greater loss after a higher dose [66] following damaging exercise. They do not establish an individual safe threshold. Parr [54] found impaired muscle protein synthesis after a high dose; its effect magnitude cannot be assigned to every event. Normal strength alone does not prove unchanged hypertrophy recovery. **PE**.

Usual Saturday event: normal D stays Friday if ready. In test week, Saturday bench may precede a known limited event only while sober and fully ready. For a larger/unfamiliar event that day, defer bench to the next eligible day after it; use verification return and page-20 spacing, shifting the next B if needed. No formal test-week Saturday aerobics. Whole-session deferral follows page 23; omitted accessories create no debt. Late Friday events use the same rules.

Bulking, cutting and substitutions

Only training changes are specified. Failure effort stays the same in both conditions; tolerate less work by reducing sets/exercises, not by covertly prescribing RIR. **PE/RI**; [17,18,42,43,49,50].

Variable	Bulking	Cutting
Olympic practice	Use phase progression and quality gates.	Initially keep frequency and secure intensity. Reduce reps/loads only when technical quality or recovery requires it; no automatic loss of skill practice.
Conventional work	Base sets, then one controlled dose trial at a time. More availability does not establish more recoverable volume.	Initially retain tolerated work. If recovery deteriorates, remove optional aerobic/athletic extras first when responsible, then recent/local hypertrophy additions. Use the targeted reduction on page 18 when broad fatigue warrants it; retain useful squat support and lagging-group work wherever possible.
Squat / bench	Progress failure rep maxima under page 20.	Do not force PRs. Keep useful load ranges and two bench days when safely possible; omit/defer an unsafe set rather than performing a long light-load failure set.
Exercise selection	Stable, familiar movements; zero glute isolation.	Same core choices. Do not switch to fatigue circuits or add conditioning solely to accelerate weight loss.
Interpretation	Progress in reps/load and physique can justify retaining a dose; it does not prove optimality.	Maintaining performance can be useful. No promise that failure preserves all muscle or that a deficit leaves your Olympic performance unchanged.

The 2026 cutting trial [17] found no additional fat-free-mass retention from 30 versus 12 weekly sets in 16 trained participants over four weeks, but some strength outcomes favored more volume. Together with earlier work [42,43], this supports individual adjustment rather than mandatory large reductions or maximum volume. **PE**.

Unavailable / unsuitable	Replacement rule - RI
Incline machine	Incline Smith with safeties, or DB with a safe endpoint; same sets, reps and tempo. Comfortable 30-45 degrees; no proven optimal angle.
Cable lateral / supported row	DB lateral raise / another chest-supported machine row; same failure endpoint.
Seated curl / calf / crunch machine	Lying leg curl / supported knee-extended calf press / cable crunch; same dose and safe finish. Calf, triceps and leg-extension setup exceptions: page 34.
No bench or squat protection	Defer the barbell failure set until proper protection is available. Machine/DB pressing does not fulfill the flat BARBELL bench requirement.
Painful movement	Stop the affected exercise; do not pick a substitute solely to keep pushing through pain. Use the readiness rules before resuming.

Run sheet and requirement audit

Record	What to log
Before	Phase/session; green/amber/red; sport/alcohol disruption; local soreness; actual time of last bench exposure.
Olympic work	Loads, successful/failed attempts, good-rep proportion, best secure single and one recurring technique issue.
Failure work	Exercise/load; valid reps in each set; endpoint: strict-form 0 RIR, unsuccessful concentric attempt, TECH, or pain/other stop. Do not record a safety stop as muscular failure.
After / next exposure	Changes made, next-session receiving/lockout quality, unusual soreness and whether it alters movement.
Every normal Friday	Compare like-load Olympic quality and first-set failure performance. Decide hold, one trial addition, reversal or reduced week.
Every pivot	Valid total or technical benchmark; lift videos; squat/bench rep performance; standardized physique photos/circumferences; optional CMJ/short-run trend from eligible athletic sessions.

A short decision sequence

Apply event and safety rules. Warm up. Complete the next priority Olympic session with its phase dose. Perform eligible conventional work at the stated failure endpoint. Do only the lower-priority work that passes readiness. Log the result, check the next bench slot and continue the sequence. No missed-volume debt. **RI.**

Original requirement	Where this edition handles it
Assumptions / evidence labels	Pages 1-3; S/PE/RI distinguish general findings, partial applicability and exact choices. Pages 27-33 add progression, monitoring, assistance and evidence detail.
Year / every session / skill / warm-ups	Pages 4-14 and 16; phase exceptions 18-19; independent component assessment 27-28.
Lagging muscles / glutes	Pages 13-15; 6 incline and 8 lateral sets at base, with measured expansion; zero glute isolation.
Athleticism / longevity / bulk-cut	Pages 21-22, 25 and 29; no nutrition, sleep or supplement instructions.
Fatigue / deload / progression / tests	Pages 17-20, 27-28 and 30; targeted reductions retain failure on every performed conventional work set.
Pickup / alcohol	Pages 23-24; day-of, day-after and rescheduling rules.
Two flat-bench days / one low-rep	Pages 11-12 and 18-20: one 3-5-rep-target failure slot, one 6-8-rep-target failure slot in every planned phase.
Unresolved scientific limitation	No study establishes the optimal year-long combination or proves failure has no cost to weightlifting. The program manages that constraint without claiming otherwise.

Independent clean and rack-jerk references

A 205-lb clean & jerk establishes one successful clean and jerk at that load; it does not reveal either separate maximum. A 255-lb power clean does not establish a full-clean maximum. Component testing can help distinguish receiving, clean recovery and jerk limitations, without inventing a strength ratio diagnosis [44,45,61].

Step	Bounded prescription
Eligibility	After two secure normal Olympic weeks, safe release competence and normal readiness. Assess in F/B or its green review week, replacing work, never after an extra max workout. This is an exception to held review-week sets/loads (pages 6/18). Assessment singles alone may reach effort 8, overriding ordinary F caps (pages 7/10/11/28). Other work keeps its phase cap. No assessment in R, pivot, fatigue reduction or taper.
Full clean: one Tuesday B	Warm up normally. Replace the six CJ work pairs with three CJ pairs at 60-70% CJ, then up to three full-clean singles. Start at a known secure clean load, usually current CJ or less. Add 5-10 lb only after an easy secure stand-up. Rest 3-4 min. Stop at technical effort 8 or the first material error; no retry. Keep the ordinary hang-snatch slot only if green.
Rack jerk: next eligible Thursday C	Replace its rack-jerk row. This assessment ramp replaces page 9: light bar, then 40/60/75% of intended first single for 3/2/1 reps. First work single about 80% CJ or a lower secure load; up to five work singles, capped at normal planned jerk reps, 3-4 min rest. After effort <=6 add 10-20 lb; at 7 add 5-10 lb; at 8 stop. Stop at first miss, press-out or unstable recovery, without retry.
Record references	CL = heaviest valid full-clean assessment single; RJ = heaviest valid rack jerk. They are demonstrated technical references, not 1RMs or proven maxima. Record date, effort and video. SN/CJ retain their own anchors; never replace CJ with CL or RJ.
Before assessment is eligible	Thursday rack jerk: 3 x 2 at a secure 50-70% CJ, effort <=6, rest 2-3 min. This is temporary teaching work. Use the page-16 regression if even that is insecure. No upward loading merely to fill the table.
Reassessment	Review CL/RJ at pivots. Repeat the bounded replacement only when it will resolve a limitation or references have become stale, normally once per cycle. Repeat sooner only if the result will change priority-1 programming and readiness permits. An ordinary valid higher component single can update the demonstrated reference without a separate test.

Use the observations, not a universal ratio

Clean receive/stand-up poor while rack jerks are secure: prioritize full-clean positions and useful squat support. Clean comfortably exceeds CJ while the jerk fails: prioritize the jerk pattern and its working-load progression. Both components exceed the combined lift: practice the transition and full pair under controlled fatigue. These are coaching hypotheses, not validated diagnostic cutoffs. **PE/RI**.

Once RJ exists, use page 7's starting percentages and page 28's direct progression. A technical reference deliberately below a maximum must not permanently cap training. Stop rules and secure demonstrations control progression, not the initial reported CJ number.

Heavy practice and jerk progression

Use performance to revise the heavy dose. Count every attempt above 90%, including misses and any preparation attempt; technical effort is not RIR. Progress one lift at a time while other workload changes are held.

Work	Progression and limit
Foundation D	After two secure exposures at 80-85% with normal subsequent practice, the final D single/pair may progress to 85-88%, effort ≤ 8 . In later F cycles, retain one previously secure 85-90% D exposure if recovered; do not repeatedly relearn lighter loads solely because the cycle restarted.
First heavy exposure	In B/R, after two secure 85-88% exposures, use one final D attempt at 90-92%, then 93-95% in R after two secure exposures. The other D slots remain $\leq 90\%$ until their heavy dose is earned. At most effort 9 for final D; no extra attempts after scheduled work.
Test more heavy practice	In B/R, after two green weeks at the current heavy dose, an additional heavy-practice trial is eligible without waiting for a plateau. Add only ONE $>90\%$ attempt per lift/week, replacing lighter work: first final C snatch / final B CJ; next penultimate D; then final A (a snatch double becomes one single). Added attempts 90-92%, effort ≤ 8 , rest 3-4 min.
Review and retention	Review at 2, 3 and 4 weekly heavy attempts per lift. Retain only with useful heavy execution and normal next-session quality/output. If more is justified, use the same two-week, one-replacement trial in remaining C/B singles, then other A sets; keep at least one light A set per lift. No automatic increase or lifetime cap at 2/3. No $>95\%$ outside tests.
Rack-jerk loads	Start from page 7. After two C exposures with every rep secure, add 2.5-5 lb total. Load may exceed the starting band or old RJ; F effort ≤ 7 , B/R ≤ 8 . On changing doubles to singles, begin at the low end of the next phase or last secure single. A valid higher single updates RJ; no inferred 1RM from reps.
Setback / squat support	Reverse a heavy addition if priority-1 output worsens; persistent faults use page 16. No additions in checkpoint, week 11/12 or reduced weeks. After two stable green weeks in F/B, repeated lift/video and squat-log review may justify one extra relevant squat set before a plateau: e.g., difficult clean stand-up or a plausible strength limitation in jerk drive. Neither squat nor clean must stall first. Use 3-5 reps to failure, rest 4-5 min; pages 31-32 govern the trial.

PE: middle amounts of very heavy work performed well in junior lifters [3]; this does not establish your optimum. Heavy reps replace work, with misses counted. Adequate practice matters as much as fatigue control. Page 31 distinguishes jerk drive, fixation and technique and supplies bounded assistance trials [33,44-46,61].

Athletic progression and relocation

Situation / next step	Runnable prescription
Monday missed	Use Thursday after C with the current dose and full warm-up/rests. If Monday impaired Tuesday, reduce/suspend it first. If neither slot is ready, omit; no Saturday/Sunday catch-up. One missed week does not restart familiarization.
Faster running / cuts	Once 4 x 20 m is tolerated: on alternate exposures replace the last two runs with 2 flying 10 m runs after a progressive 20 m run-in, first 90%, then 95% after two green exposures; rest 3-4 min. Alternatively replace those runs with 2 sets of one 45-degree cut each side: approach/exit 5 m, initially 75-85%, then 85-90%; rest 2 min/set, 20-30 s/side. Choose one option.
Second exposure	After four productive primary exposures, trial Thursday jumps 2 x 3; full page-21 preparation. Hold Monday dose while trialing Thursday. After two green Thursday exposures, trial 2 x 10 m runs at 85-90%, rest 2-3 min. Check Friday lifting/failure output. At most two weekly athletic slots.
Further development	Change one weekly component after two green exposures: +3 jumps, +one short run, or the next listed intensity/distance step. Review at 24-36 purposeful jumps and 6-8 short runs/week across both slots. Further additions use the same rule if useful; these are neither quotas nor proven ceilings. Do not raise all components together.
Checkpoints / R	Green weeks 4/8 retain the established frequency, dose and intensity without progression. R weeks 9-10 retain them if higher-priority work remains normal; no new drills or additions. Reduce only for observed cost or an active event restriction, not merely the phase name.
Testing / return	Week 11: Monday only, halve current jump SETS and run REPS, rounding up; keep familiar intensity, no progression. Week 12: no module. Week 13: resume the pre-taper dose if recovered, without additions. After >14 days off athletics, return one progression step lower for two good exposures before restoring.
Pickup / measurement	A demanding game replaces the nearest overlapping module, secondary first. A possible game does not cancel planned work. Record purposeful jumps, monitoring jumps and low-intensity warm-up contacts separately; monitoring reps count within the planned jump work, never as free extra volume.

Review athletic trends every pivot using existing jumps/timed runs; testing replaces those reps. A 20 m run-in does not guarantee maximal speed, and planned cuts are not full reactive agility. Evidence for training benefits is broader than these exact steps [21,22,27,63].

Preserve productive Olympic and hypertrophy work first. If health-only conditioning impairs athletics, reverse that addition. If athletics itself impairs either higher priority, reverse its responsible dose. Retained development should persist across cycles; it is not automatically reduced to the entry dose.

Monitoring and annual volume audit

Measure	How to use it
Olympic output	Use consistent video angles and familiar-load warm-ups. Log valid reps, misses, best secure load and technical effort. Compare like phases/loads, not the heaviest R single against an easy F double. Check Monday/Thursday after failure days and Tuesday/Friday after athletics.
Conventional output	Log load/reps, endpoint and visit position. Compare first sets at the same setup/load; >20% loss on two comparable exposures prompts review. Later-session decline uses the two-exposure split trial on page 4. Address obvious impairment immediately.
Optional objective jump measure	If reliable jump measurement is already available, record the mean of three identical CMJs within the planned jump dose. Same device, surface, arm use and warm-up. Establish at least three normal-session values before interpreting trends. A persistent decline beyond usual variation is a review flag, not automatic clearance or a diagnostic cutoff [62].
Physique / support	At each pivot use consistent photos and circumferences; review upper chest, side delts and traps with the load/rep log. No short-term tape change proves extra hypertrophy. Preserve useful strength support even when glutes need no extra isolation.
Two-week dose review	Choose hold, one additional weekly set or athletic/aerobic step, reversal, or a targeted reduction. Resolve priority-1 needs first. A tolerable dose is not necessarily optimal; hold other changes while testing the next one.

What the year actually contains

Normal base: 22 conventional failure sets per Tuesday/Friday (preferred split 9 + 13), 44 per week. This includes 6 incline, 8 lateral, 2 direct shrug, 2 squat and 2 flat-bench sets weekly. Successful dose trials alter those totals. Olympic sets are tracked separately, without invented hypertrophy-set credits. **RI.**

With green checkpoints/pivots and no additions or disruptions, an established 13-week cycle has 522 conventional sets: eleven 44-set weeks, a 36-set week 11, and a 2-set test week. First-cycle onboarding lowers this to 498. Four cycles total 2,064 sets, averaging about 39.7/week. These are transparent planned counts, not targets to repay after omissions. **RI.**

Under the same assumptions, annual direct incline/lateral/shrug averages are about 5.3 / 7.0 / 1.85 sets per week before successful additions; both squat exposures occur in 48 of 52 weeks. The actual log is the final audit when events, recovery or substitutions change a dose. **RI.**

Every performed conventional work set uses the confirmed failure endpoint; Olympic work, preparation and athletics remain quality-limited. Every planned week contains separate low-rep and moderate flat-BARBELL bench slots. Unsafe readiness can defer a slot, never justify forced reps or compressed catch-up days. All four goals remain ordered; no research proves this exact combination is globally optimal.

Choose assistance for the observed limit

Review at least three comparable lifting exposures and relevant component/squat observations. These support a working hypothesis, not a diagnosis from ratios. Full and rack jerks already train loaded dip and drive. Choose one useful support trial; no plateau is required. New trials start in normal F/B after two stable green weeks, with other dose changes held [33,44,45,61].

Observed pattern	First response / bounded trial
Dip drift, unstable catch or inconsistent timing	Use the lighter page-16 technique regression first. A forward drive or missed footwork does not prove weak pressing. Keep secure floor-lift practice and one cue.
Drive needs support; positions repeatable	First progress rack-jerk load under page 28. For additional dip-drive rehearsal, replace the first two C jerk sets with 2 x 1 pause-dip split jerk at 60-75% assessed RJ: pause 1 s at dip bottom, drive and receive normally; rest 3 min, effort ≤ 7 . Remaining C sets use their approved load. Without RJ or stable technique, use pages 16/27 instead. Quality-limited, never failure.
Additional squat strength plausibly useful	Use page 28: one extra front- or back-squat work set, 3-5 valid reps to failure, 4-5 min rest, after that day's existing squat and before bench. Choose the variation most relevant to the observed stand-up/drive task. Both lifts need not stall. No extra hinge/glute isolation.
Fixation strength plausibly useful; position stable	A trial may precede repeated misses when component/video observations suggest limited overhead strength. Seated machine or supported DB press: 1 x 6-10 to failure Friday after flat bench, before incline. Replace ONE Friday incline set. Warm up about 40% work load x 8, 65% x 4; rest 3-4 min; 2 s lowering. No invented maximum.
Several patterns / unclear limitation	Choose the best-supported hypothesis and one trial. Do not add presses, squats and jerk variations together. If uncertainty persists, seek coaching review. A different assistance exercise needs its own clear target, dose and failure/quality endpoint before substitution.

Start a trial with a target and a record

Record the exercise, start date, baseline comparison and weekly set change. For squat support, track a familiar-load clean stand-up/jerk drive and squat rep-load performance; for overhead work, track valid jerks and fixation at familiar loads plus assistance performance. Keep camera angle, setup and session position comparable. Do not add max tests to manufacture a result. Page 32 separates tolerance checks from later benefit review.

Press/jerk maxima correlate [61]; training transfer is unproved. Seated pressing has no demonstrated recovery or jerk-transfer advantage over standing pressing. Its controllable failure endpoint motivates this trial (RI); pause jerks remain quality-limited. Review snatch pulls by the same target-and-cost standard on page 34.

Assistance trial reviews and continuity

A short trial can reveal poor tolerance or a better movement pattern. It cannot establish the absence of long-term strength transfer. The following review windows are practical decisions, not validated diagnostic thresholds [2,3,33,44,45,61].

Stage / circumstance	Action
After each exposure	Record the target measures from page 31 and the next relevant Olympic session. Pain, clear interference or worsening positions ends the trial: remove an added squat set, restore the replaced incline set, or return to regular C jerks at a secure load. Restore work only if locally ready; page 17 overrides.
After 2 exposures: tolerance	Check next-session output, local positions and ability to perform the intended assistance. If tolerated, keep the same set dose. Progress load only under the rule below; do not add another exercise or more sets.
After 4 exposures: early direction	Review target movement and rep-load trends. Clear useful improvement may justify retention. If tolerated but transfer is uncertain, continue to 8 eligible exposures; lack of a quick jerk PR is not failure. Hold other planned dose increases through this window where practical; necessary changes take priority and make attribution uncertain.
At 8 exposures: benefit review	Compare repeated target/strength observations and opportunity cost. Retain a useful trial without priority-1 loss. If tolerated and promising but transfer unclear, specify one further 4-exposure review with the same dose, or remove/restore the replaced work. Record why. No automatic expiry or proof of failure from absent short-term transfer.
Green checkpoints and R weeks 9-10	Existing tolerated trials may continue at the same set dose and most recent secure load even before the benefit review is complete. Do not start or escalate a trial. Pause-jerk replacements may remain only while regular full/rack jerk quality is normal; keep the remaining C sets as ordinary split jerks.
Week 11	Return pause-jerk replacements to ordinary C split jerks. Tuesday retains tolerated conventional support; Friday uses one set per exercise, so no extra squat set. An active overhead press uses one Friday set in place of the one incline set; no Friday incline in that case.
Taper, pivot and interruptions	Week 12: no assistance. Week 13: base prescriptions, review the log; resume any still-promising trial next F when ready. Reduced/event restrictions override continuation. Count only completed comparable exposures; preserve the log across a pause rather than restart the trial. After a break, use a secure load and repeat the tolerance check before progressing.

Load progression: in normal F/B, the extra squat/press set follows page 20. For pause jerks, add 2.5-5 lb after two secure exposures, within 60-75% RJ and effort ≤ 7 . Keep the same number of sets. Checkpoints and R hold the last secure trial load unless a reduction is needed. A useful retained trial keeps these phase rules. Record every substitution; base audit totals exclude trials.

Each normal failure day has a 9 + 13 base-set split. An extra squat set adds one to the first visit on its day. A press substitution keeps that visit's total unchanged while removing one incline set; a pause-jerk substitution changes Olympic reps, not conventional sets. Two separate flat-bench days, including the low-rep slot, remain prescribed.

Evidence findings and their application

Labels describe the stated claim, not a paper or the whole plan. Directness and certainty differ: a small direct trial is not automatically strong evidence for every lifter. Page 3 discusses conflicting findings; this table separates selection evidence from the exact prescription.

Decision	What the research supports	What remains inferred here
Weekly resistance dose	S: resistance training builds strength/size [1]; higher volume generally adds benefit with diminishing returns [8].	RI: 44 weekly failure sets, allocation, 9/13 split and one-set trials. No precise hypertrophy credit for each Olympic set.
Lateral raises	PE / direct trial: 24 trained adults, 8 weeks, 8-10 weekly sets to momentary failure; cable and DB produced similar growth [12].	RI: four base sets per session and progression. Failure was studied directly; the combined weightlifting workload was not.
Seated leg curl	PE / direct trial: seated training grew hamstrings more than prone over 12 weeks in 20 adults without recent systematic training [39].	RI: two sets twice weekly at failure in this trained athlete. Selection has direct support; the exact dose is extrapolated.
Incline / traps / glutes	PE: incline study used untrained men and a 44-degree Smith press [13]; squats can grow glutes [40]. Shrug-variant hypertrophy evidence is limited.	RI: comfortable 30-45-degree machine incline, supported shrug, no glute isolation. Useful squat support stays ahead of physique preference.
Olympic loading / assistance	PE: moderate total and heavy volumes performed well in junior lifters [2,3]; press/jerk correlation [61] and weightlifting statement [33] inform support.	RI: exact phase loads, heavy replacements, component assessments and assistance trials. More practice must earn retention, not just more fatigue.
Athletics / mobility	S: plyometric training improves athletic outcomes on average [21]; resistance training/stretching improve ROM [23,24]. PE: priming [64]; microdose study [63] has inconsistent dose reporting.	RI: entry, increments, timing and return rules. Warm-up status does not make extra jumps free; more mobility is not automatically useful.
Conditioning / events	PE: concurrent evidence differs by outcome/context [19,20]; alcohol and sport studies use specific exposures [29,30,54,65-68].	RI: progressive aerobic dose, demand-based game changes, verification return. Readiness does not prove unchanged long-term adaptation.
Failure / deloads	PE: failure comparisons differ in definitions, doses and outcomes [9,10,48-53,55-60]. Deload consensus and trials do not establish one optimal schedule [36-38].	RI: user-selected failure, adaptive checkpoints, targeted reductions and taper. No lighter high-rep failure substitute for reducing workload.

Targeted research through 13 September 2026; publication dates can precede issue dates. New direct exercise comparisons and their implementation limits are on page 34. Verified source identity, tolerability and arithmetic correctness do not establish an optimal individual prescription.

Exercise comparisons and trial options

Selection / comparison	Implementation and evidence boundary
Overhead cable extension vs pressdown	Default: page 13 overhead extension. Direct MRI evidence favors it [69], but used 21 adults without recent training and a higher dose. Pressdown, same 1 x 10-15, is a fallback for actual intolerance or repeated observed rack/overhead interference; no assumed recovery advantage.
Standing vs seated calves	Both days now use knee-extended raises [70]. If shoulder loading is unsuitable, use a supported knee-extended calf press. Seated is an individualized fallback; it is not required for greater soleus growth.
Reclined leg extension	Prefer about 40 degrees anatomical hip flexion: 0 = straight hip; upright sitting about 90. This opens the interior torso-thigh angle toward 140 degrees [71]. Keep back/pelvis supported and knee axis aligned. Use normal seated extension if the machine cannot do this securely; do not imitate it with spinal arching.
Optional calf ROM trial	After two stable green F/B weeks at a familiar standing dose, replace ONE Friday calf set with initial partials: heel-down dorsiflexion to neutral ankle, 10-15 reps to failure of that fixed range; 2 s down, rest 2-3 min. Hold other changes; check tolerance twice, execution/rep trends at 4/8 exposures and physique over a full cycle. Same sets; no appended finishers. Results [75-77] justify consideration, not this exact mixed dose. Phase rules: page 32.
Incline / cable curl	Use a consistent comfortable 30-45-degree incline; 30 is not a proven optimum [13]. Keep neutral-shoulder cable curl: the matched 2026 comparison does not establish an extended-shoulder advantage [73]. Neither choice proves this set dose optimal.
Snatch-pull purpose and review	The base C pull is provisional position/extension rehearsal, not guaranteed strength overload from a skill-limited SN anchor. At the first green checkpoint, record its target. If none is apparent, omit it for the next two C exposures, hold other changes, and compare full snatch execution. Retain omission if no useful loss; restore only with a plausible target and normal readiness. No replacement volume is owed.
Volume and calendar	One-set failure can work [51]; it does not establish maximum gains. Higher failure-squat volume sometimes helped [74]; a large volume increase did not clearly help or harm measured growth in another trained cohort [79]. Several WL loading structures work [78]. Base sets, small increments and four annual peaks remain provisional choices, not proven optima.

No direct ranking was established for supported rows, pulldowns, shrug variants, reverse pec deck or machine crunch. Keep them for their stated task and controllable execution, and revise when a better-supported option or individual response warrants it. Absence of a located comparison is not evidence of equality.

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Sources underlying this edition and the retained literature review. A citation supports the stated finding, not every detail of an exact RI prescription.

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